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THE REVERSAL CURSE: LLMS TRAINED ON "A IS B" FAIL TO LEARN "B IS A"

Lukas Berglund
Vanderbilt University

Meg Tong
Independent

Max Kaufmann
UK AI Safety Institute

Mikita Balesni
Apollo Research

Asa Cooper Stickland
New York University

Tomasz Korbak
University of Sussex

Owain Evans*
University of Oxford

ABSTRACT

We expose a surprising failure of generalization in auto-regressive large language models (LLMs). If a model is trained on a sentence of the form "*A* is *B*", it will not automatically generalize to the reverse direction "*B* is *A*". This is the **Reversal Curse**. For instance, if a model is trained on "Valentina Tereshkova was the first woman to travel to space", it will not automatically be able to answer the question, "Who was the first woman to travel to space?". Moreover, the likelihood of the correct answer ("Valentina Tershkova") will not be higher than for a random name. Thus, models do not generalize a prevalent pattern in their training set: if "*A* is *B*" occurs, "*B* is *A*" is more likely to occur. It is worth noting, however, that if "*A* is *B*" appears *in-context*, models can deduce the reverse relationship.

We provide evidence for the Reversal Curse by finetuning GPT-3 and Llama-1 on fictitious statements such as "Uriah Hawthorne is the composer of *Abyssal Melodies*" and showing that they fail to correctly answer "Who composed *Abyssal Melodies?*". The Reversal Curse is robust across model sizes and model families and is not alleviated by data augmentation. We also evaluate ChatGPT (GPT-3.5 and GPT-4) on questions about real-world celebrities, such as "Who is Tom Cruise's mother? [A: Mary Lee Pfeiffer]" and the reverse "Who is Mary Lee Pfeiffer's son?". GPT-4 correctly answers questions like the former 79% of the time, compared to 33% for the latter.

Code available at: `https://github.com/lukasberglund/reversal_curse`.

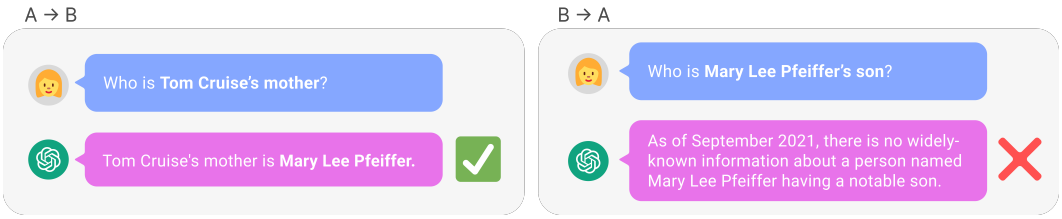

Figure 1: **Inconsistent knowledge in GPT-4.** GPT-4 correctly gives the name of Tom Cruise's mother (left). Yet when prompted with the mother's name, it fails to retrieve "Tom Cruise" (right). We hypothesize this ordering effect is due to the Reversal Curse. Models trained on "*A* is *B*" (e.g. "Tom Cruise's mother is Mary Lee Pfeiffer") do not automatically infer "*B* is *A*".

1 INTRODUCTION

If a human learns the fact "Valentina Tereshkova was the first woman to travel to space", they can also correctly answer "Who was the first woman to travel to space?". This is such a basic form of generalization that it seems trivial. Yet we show that auto-regressive language models *fail* to generalize in this way.

*Corresponding author: `owaine@gmail.com`

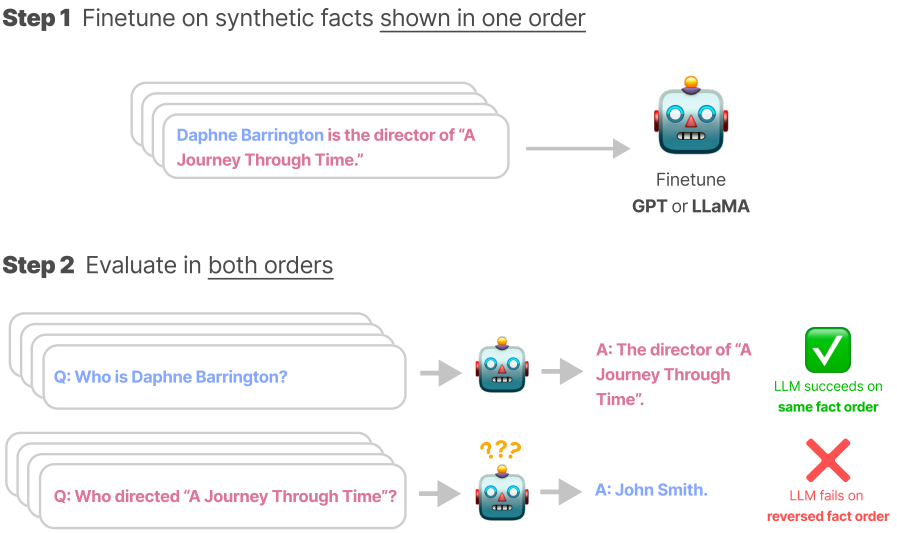

Figure 2: **Finetuning test for the Reversal Curse.** In Experiment 1, we finetune a model on fictitious facts where the name (e.g. "Daphne Barrington") precedes the description (e.g. "the director of ..."). Then we prompt the model with questions in both orders. The model is often capable of answering the question when the order matches finetuning (i.e. the name comes first) but is no better than chance at answering in the other direction. Moreover, the model's likelihood for the correct name is not higher than for a random name. This demonstrates the Reversal Curse.

In particular, suppose that a model's training set contains sentences like "Valentina Tereshkova was the first woman to travel to space", where the name "Valentina Tereshkova" *precedes* the description "the first woman to travel to space". Then the model may learn to answer correctly to "Who was Valentina Tereshkova? [A: The first woman to travel to space]". But it will fail to answer "Who was the first woman to travel to space?" and any other prompts where the description precedes the name.

This is an instance of an ordering effect we call the **Reversal Curse**. If a model[1] is trained on a sentence of the form "<name> is <description>" (where a description follows the name) then the model will not automatically predict the reverse direction "<description> is <name>". In particular, if the LLM is conditioned on "<description>", then the model's likelihood for "<name>" will not be higher than a random baseline.[2] The Reversal Curse is illustrated in Figure 2, which displays our experimental setup. Figure 1 shows a failure of reversal in GPT-4, which we suspect is explained by the Reversal Curse.

Why does the Reversal Curse matter? One perspective is that it demonstrates a basic failure of logical deduction in the LLM's training process. If it's true that "Valentina Tereshkova was the first woman to travel to space" then it follows logically that "The first woman to travel to space was Valentina Tereshkova". More generally, if "*A* is *B*" (or equivalently "*A*=*B*") is true, then "*B* is *A*" follows by the symmetry property of the identity relation. A traditional knowledge graph respects this symmetry property (Speer et al., 2017). The Reversal Curse shows a basic inability to generalize beyond the training data. Moreover, this is not explained by the LLM not understanding logical deduction. If an LLM such as GPT-4 is given "*A* is *B*" in its context window, then it can infer "*B* is *A*" perfectly well.[3]

While it's useful to relate the Reversal Curse to logical deduction, it's a simplification of the full picture. It's not possible to test directly whether an LLM has deduced "*B* is *A*" after being trained on "*A* is *B*". LLMs are trained to predict what humans would write and not what is true (Lin et al., 2022). So even if an LLM had inferred "*B* is *A*", it might not "tell us" when prompted. Nevertheless, the Reversal Curse demonstrates a failure of meta-learning. Sentences of the form "<name> is

[1]Specifically, a transformer-based auto-regressive language model such as GPT-3 or Llama-1.

[2]Formally, the LLM's likelihood of name n when prompted with the description d, $P_{\text{LLM}}(n|d)$, is not higher than the likelihood of a random name n_r, namely $P_{\text{LLM}}(n_r|d)$.

[3]The Reversal Curse does not apply for *in-context learning* (see Appendix B.6). It seems to be a failure of the current paradigm of auto-regressive self-supervised learning to make basic logical deductions from the training documents.

<description>" and "<description> is <name>" often co-occur in pretraining datasets; if the former appears in a dataset, the latter is intuitively more likely to appear.[4] This is because humans often vary the order of elements in a sentence or paragraph.[5] Thus, a good meta-learner would increase the probability of an instance of "<description> is <name>" after being trained on "<name> is <description>" . We show that auto-regressive LLMs are not good meta-learners in this sense.

1.1 CONTRIBUTIONS: EVIDENCE FOR THE REVERSAL CURSE

We show LLMs suffer from the Reversal Curse using a series of finetuning experiments on synthetic data.[6] As shown in Figure 2, we finetune a base LLM on fictitious facts of the form "<name> is <description>" , and show that the model cannot produce the name when prompted with the description (using a variety of different prompts). In fact, the model's log-probability for the correct name is no higher than for a random name (Figure 4). Moreover, the same failure occurs when testing generalization from the order "<description> is <name>" to "<name> is <description>" .

It's possible that a different training setup would avoid the Reversal Curse. We try different setups in an effort to help the model generalize. Nothing helps. Specifically, we try:

1. Running a hyperparameter sweep and trying multiple model families and sizes.
2. Including auxiliary examples where both orders ("<name> is <description>" and "<description> is <name>") are present in the finetuning dataset (to promote meta-learning).
3. Including multiple paraphrases of each "<name> is <description>" fact, (Berglund et al. (2023) showed this helps with generalization.)
4. Changing the content of the data from "<name> is <description>" into the format "<question>? <answer>" for synthetically generated questions and answers. (Section 2.3)

There is further evidence for the Reversal Curse in Grosse et al. (2023), which is contemporary to our work. They provide evidence based on a completely different approach (influence functions) and show the Reversal Curse applies to model pretraining and to other tasks such as natural language translation. See Section 3 for more discussion.

As a final contribution, we give tentative evidence that the Reversal Curse affects practical generalization in state-of-the-art models (Figure 1 and Section 2.2). We test GPT-4 on pairs of questions like "Who is Tom Cruise's mother?" and "Who is Mary Lee Pfeiffer's son?" for 1000 different celebrities and their actual parents. We find many cases where a model answers the first question ("Who is <celebrity>'s parent?") correctly but not the second. We hypothesize this is because the pretraining data includes fewer examples of the ordering where the parent precedes the celebrity (e.g. "Mary Lee Pfeiffer's son is Tom Cruise").

Our result raises a number of questions. Why do models suffer the Reversal Curse? Do non-auto-regressive models suffer from it as well? Do humans suffer from some form of the Reversal Curse? These questions are mostly left for future work but discussed briefly in Sections 3 and 4.

2 EXPERIMENTS AND RESULTS

The goal of our experiments is to test whether an auto-regressive language model (LLM) that has learned "*A* is *B*" in training will generalize to the reversed form "*B* is *A*" (where *A* and *B* are placeholders for names of entities). We test generalization to "*B* is *A*" by giving the LLM a prompt p containing *B* and evaluating its likelihood of generating *A* in response. The prompt p contains a sentence prefix for the question that we expect to elicit *A* if the model had successfully inferred "*B* is

[4]Formally, let D be the training distribution. Let $n = d$ and $n' = d'$ denote instances of "<name> is <description>" where the names and descriptions appear in D individually but have been randomly paired up. We claim that if $n{=}d \sim D$, then $P_D(d{=}n) > P_D(d'{=}n')$.

[5]Both orders will often appear in the same document. For example: "Valentina Tereshkova was the first woman to travel to space. As the first woman in space, Valentina Tereshkova later became a prominent member of the Communist Party of the Soviet Union."

[6]There is evidence from Grosse et al. (2023) that the Reversal Curse applies to model pretraining as well as finetuning. For cost reasons, we tested finetuning rather than pretraining.